AOU
الجامعة العربية المفتوحة
Arab Open University
Faculty of Computer Studies

MT132 – M132
Linear Algebra

Final Examination
Spring Semester 2024/2025

Date: 2 /June/2025

Number of Exam Pages: 3 pages (including this cover sheet)	**Time Allowed:** 2 Hours

Instructions:

- Total Marks: **50**
- This exam consists of **3** parts.
- **ALL questions** must be answered in the External Answer booklet.
- Be sure you write your **name and ID** on the External Answer booklet.
- **Calculators** are allowed.

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Q–1: [5×3 marks] Choose the correct answer:

1.1: The inverse of $\begin{bmatrix} 1 & -1 \\ 2 & 2 \end{bmatrix}$

a) $\frac{-1}{4}\begin{bmatrix} 2 & 1 \\ -2 & 1 \end{bmatrix}$

b) $\frac{1}{4}\begin{bmatrix} 2 & 1 \\ -2 & 1 \end{bmatrix}$

c) -4

d) 4

e) None of the above

1.2: If $W = Span\left\{\begin{bmatrix} -1 \\ 3 \end{bmatrix}, \begin{bmatrix} 3 \\ -9 \end{bmatrix}, \begin{bmatrix} 3 \\ 9 \end{bmatrix}\right\}$, then $\dim(W) =$

a) 0

b) 1

c) 2

d) 3

e) None of the above

1.3: Suppose that X_1, X_2 and X_3 are solutions for the linear system $AX = \mathrm{B}$. $5kX_1 - k^2X_2 - 5X_3$ is also solution only if

a) $k = -2, -3$

b) $k = 2, 3$

c) $k \neq -2, -3$

d) $k \neq 2, 3$

e) None of the above

1.4: The matrix $\begin{bmatrix} 2 & k \\ 2 & -3 \end{bmatrix}$ is singular if

a) $k = 3$

b) $k = -3$

c) $k \neq 3$

d) $k \neq -3$

e) None of the above

1.5: The vectors $v_1 = \begin{bmatrix} k^2 \\ 2k \end{bmatrix}, and\ v_2 = \begin{bmatrix} 16 \\ 4k^2 \end{bmatrix}$ are linearly independent if

a) $k = 0, 2$

b) $k = 0, 8$

c) $k \neq 0, 2$

d) $k \neq 0, 8$

e) None of the above

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Q–2: [8 marks] Answer the following questions as True or False (2 mark), Justifying your answer (2 marks):

2.1: The linear system $\left[\begin{array}{cc|c} 5 & -4 & -2 \\ 1 & 2 & 1 \end{array}\right]$ has no solution.

2.2: If $|A - B| = 0$ then $A = B$.

Each question is worth 9 marks. Answer the following questions:

Q–3: [5+4 marks]

Let $W = \left\{ \begin{bmatrix} 3a - b + 2c \\ 4b + 4c \\ a + 2b + 3c \end{bmatrix} ; a, b \in R \right\}$ be a subset of $\boldsymbol{R^3}$.

a) Show that W is a subspace of $\boldsymbol{R^3}$.
b) Find the basis and the dimension of W.

Q–4: [3+4+2 marks]

Define the linear transformation $T: R^3 \to R^2$ by $T\begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} x + 2y \\ 2y - z \end{bmatrix}$,

a) Find a matrix A such that $T(X) = AX$, for each $X \in R^3$.
b) Find the null of T, $N(T)$ and range of T, $R(T)$.
c) Find $Dim(N(T))$ and $Dim(R(T))$.

Q–5: [9 marks]

Find the eigenvalues of the matrix $A = \begin{bmatrix} -1 & 2 \\ 3 & 4 \end{bmatrix}$; then find matrices P and D such that $A = PDP^{-1}$.

End of questions

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